LAB 7 – FUZZ TESTING FOR INPUT VALIDATION BUGS

Project: L6_L7_29 – Smart Thermostat Controller

1. OBJECTIVE

The objective of this lab is to understand and implement fuzz testing as a software testing technique using the Hypothesis library in Python.

The experiment aims to identify input-validation bugs by automatically generating random, unexpected and malformed inputs for the Smart Thermostat Controller application.

The three functions tested are:

1. parse_setpoint() – Converts temperature setpoints from Fahrenheit or Celsius into Celsius.
2. average_temperature() – Calculates the average of multiple temperature readings.
3. schedule_slot() – Identifies whether a given time belongs to the night or day schedule.

The identified bugs are fixed using defensive programming techniques to improve application robustness and prevent unexpected crashes**.**

2. SOFTWARE REQUIREMENTS

The following software and libraries were used:

- Python 3.12.10
- Hypothesis 6.170.15
- Pytest 8.3.4
- Visual Studio Code
- GitHub

SCREENSHOTS

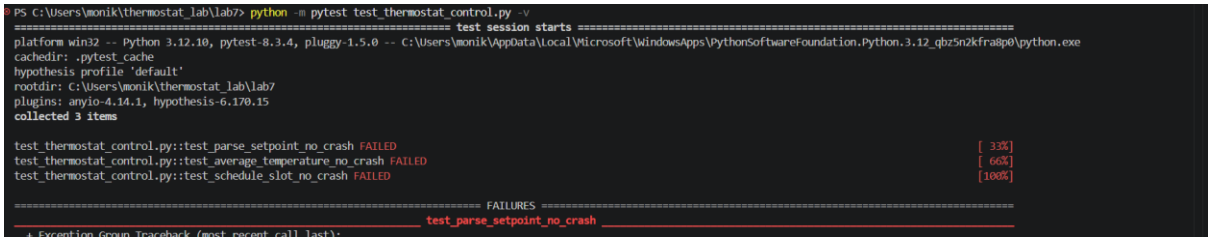
Original Application Execution – Screenshot 1

```
PS C:\Users\monik\thermostat_lab\lab7> python -m pytest --version
pytest 8.3.4
PS C:\Users\monik\thermostat_lab\lab7> python thermostat_control.py
==== Running parse_setpoint ====
Enter a setpoint (e.g. 21.5C or 70F): 70F
Setpoint in Celsius: 21.1

==== Running average_temperature ====
Enter sensor readings as CSV (e.g. 20.5,21,22.5): 20.5,21,22.5
Average Temperature: 21.333333333333332

==== Running schedule_slot ====
Enter a time as HH:MM (e.g. 07:30): 07:30
Schedule Slot: day 30
PS C:\Users\monik\thermostat_lab\lab7>
```

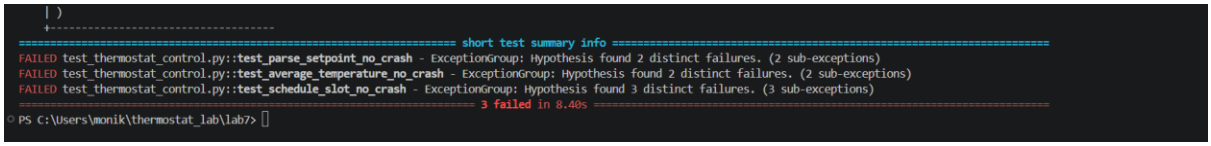
Fuzz Testing Failures – Screenshot 2



```
PS C:\Users\monik\thermostat_lab\lab7> python -m pytest test_thermostat_control.py -v
============================= test session starts ==============================
platform win32 -- Python 3.12.10, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\monik\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\monik\thermostat_lab\lab7
plugins: anyio-4.14.1, hypothesis-6.170.15
collected 3 items

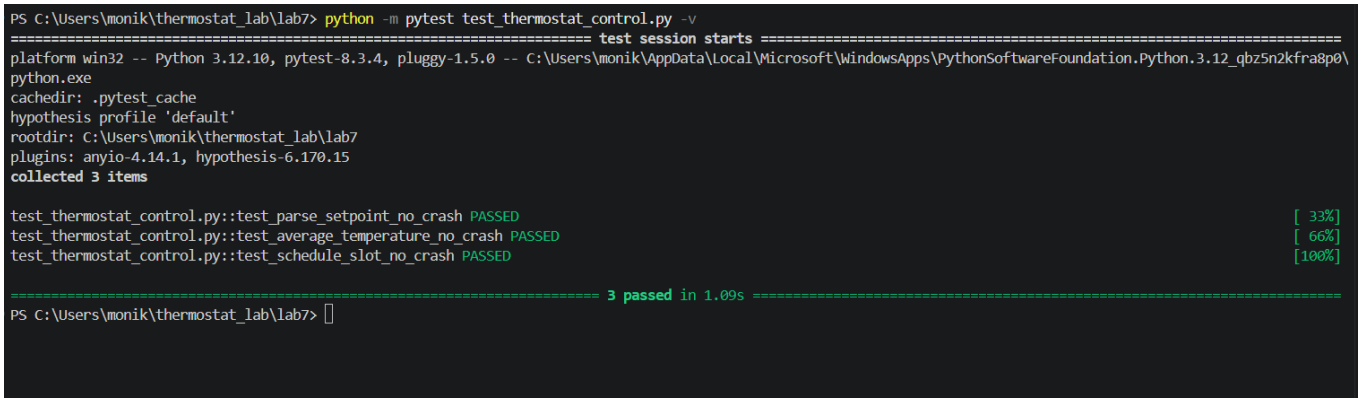
test_thermostat_control.py::test_parse_setpoint_no_crash FAILED          [ 33%]
test_thermostat_control.py::test_average_temperature_no_crash FAILED     [ 66%]
test_thermostat_control.py::test_schedule_slot_no_crash FAILED           [100%]

================================== FAILURES ===================================
_____________________ test_parse_setpoint_no_crash _____________________
  + Exception Group Traceback (most recent call last):
```



```
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_thermostat_control.py::test_parse_setpoint_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_thermostat_control.py::test_average_temperature_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_thermostat_control.py::test_schedule_slot_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 8.40s ==============================
PS C:\Users\monik\thermostat_lab\lab7>
```

Fixed Application Execution – Screenshot 3



```
PS C:\Users\monik\thermostat_lab\lab7> python -m pytest test_thermostat_control.py -v
============================= test session starts ==============================
platform win32 -- Python 3.12.10, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\monik\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\
python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\monik\thermostat_lab\lab7
plugins: anyio-4.14.1, hypothesis-6.170.15
collected 3 items

test_thermostat_control.py::test_parse_setpoint_no_crash PASSED          [ 33%]
test_thermostat_control.py::test_average_temperature_no_crash PASSED     [ 66%]
test_thermostat_control.py::test_schedule_slot_no_crash PASSED           [100%]

============================== 3 passed in 1.09s ==============================
PS C:\Users\monik\thermostat_lab\lab7>
```

Successful Fuzz Testing – Screenshot 4

